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Does migration promote or inhibit diversification? A case study involving the dominant radiation of temperate Southern Hemisphere freshwater fishes

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Although theory predicts that dispersal has a pivotal influence on speciation and extinction rates, it can have contradictory effects on each, such that empirical quantification of its role is required. In many studies, dispersal reduction appears to promote diversification, although some comparisons of migratory and nonmigratory species suggest otherwise. We tested for a relationship between migratory status and diversification rate within the dominant radiation of temperate Southern Hemisphere freshwater fishes, the Galaxiidae. We reconstructed a molecular phylogeny comprising >95% of extant taxa, and applied State-dependent Speciation Extinction models to estimate speciation, extinction, and diversification rates. In contrast to some previous studies, we revealed higher diversification rates in nonmigratory lineages. The reduced gene flow experienced by nonmigratory galaxiids appears to have increased diversification under conditions of allopatry or local adaptation. Migratory galaxiid lineages, by contrast, are genetically homogeneous within landmasses, but may also be rarely able to diversify by colonizing other landmasses in the temperate Southern Hemisphere. Apparent contradictions among studies of dispersal-diversification relationships may be explained by the spatial context of study systems relative to species dispersal abilities, by means of the "intermediate dispersal" model; the accurate quantification of dispersal abilities will aid in the understanding of these proposed interactions.

KEY WORDS: **Anadromy, diadromy, diversification, gene flow, intermediate dispersal, river capture.**

Dispersal can be defined as the movement of individuals from their place of origin to their place of reproduction, or shifts among different locations of reproduction, and such movement can have key consequences for ecology, evolution, biodiversity, and conservation biology (Clobert et al. 2004). Dispersal influences population dynamics and species ranges, and can facilitate persistence in spatially and temporally heterogeneous environments (Hanski 1994). Additionally, dispersal influences individual fitness, the colonization of new environments, levels of inbreeding and genetic variation, and the genetic divergence of populations (Frankham and Ralls 1998; Lenormand 2002). Dispersal can thus have major impacts on the rates and spatial scales of both speciation and extinction and, consequently, overall diversification (Kisel and Barraclough 2010; Ikeda et al. 2012; Waters et al. 2020). However, estimating these impacts can be problematic, given that dispersal can influence speciation and extinction in different ways (Claramunt et al. 2012). Empirical analyses are therefore needed to untangle dispersal-diversification relationships, and such studies should preferably involve taxa that differ primarily in a logical dispersal-related trait, such as flightlessness or flight ability in birds (Claramunt et al. 2012; Campagna et al. 2019), or brooding versus broadcast spawning in marine invertebrates (Puritz et al. 2012). Ideally, the veracity of any inferences should also be tested by replication on a phylogenetically distinctive group that exhibits homologous trait variation, such as the positive effect of reduced flight on insect

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diversification (Wagner and Liebherr 1992), as also observed for birds.

Migration can be variously defined, but is often understood as predictable seasonal movement of individuals between locations, and it is widely but heterogeneously distributed across animals (Dingle and Drake 2007). Although migration does not necessitate spatial shifts in reproduction (i.e., dispersal), a positive association is assumed (Lombal et al. 2020). As with other dispersal-related traits, migration presents an interesting conundrum with respect to predicting its impact on diversification (McDowall, 1988, 2001; Winker 2000; Corush 2019). Assuming some level of straying during migration, migration should lead to the colonization of new environments, which may result in adaptation and divergence. However, such mobility can also confer high levels of gene flow, which may impede divergence of populations (Slatkin 1987; Bohonak 1999; McDowall 2001; but see Nosil 2008; Abbott et al. 2013). Conversely, resident (nonmigratory) taxa typically possess lower capacity to traverse geographic barriers, but (having done so) experience much lower rates of gene flow among populations. Concomitantly, higher levels of population genetic differentiation appear linked to high diversification rates (Harvey et al. 2017). However, diversification rates also depend on extinction, with lower extinction rates expected for migratory versus resident taxa because of their broader distributions, larger population sizes (McDowall 2007; Bloom and Lovejoy 2014), and higher genetic diversity (Allibone and Wallis 1993; Ward et al. 1994; McDowall 1999; Victoriano et al. 2020). Therefore, the broad relationship between migratory status and diversification remains unclear, and the few detailed analyses conducted to date suggest either higher (Rolland et al. 2014; Fuchs et al. 2015; Corush 2019; Gómez-Bahamón et al. 2020) or lower (Tedesco et al. 2017) diversification in migratory lineages, or no relationship with migratory status (Kennedy et al. 2016). Further research is needed to understand the intricacies of diversification rates and migratory status.

Contrasts between diadromous and freshwater-limited life histories have provided iconic examples of fish diversification (McDowall 2001; Waters and Wallis 2001). Diadromy is defined as the obligatory and predictable migration of individuals between marine and freshwater environments (Myers 1949; McDowall 1988), and is thought to enable species to exploit habitats that differ in productivity (Gross 1987; Gross et al. 1988), predation pressure, or breeding suitability (Dodson et al. 2009) at different stages of the life cycle. In reality, diadromy encompasses a range of behaviors that differ chiefly with respect to whether breeding occurs in marine (catadromy) or freshwater (anadromy) environments, or whether movement between these environments occurs predictably for some other purpose (amphidromy; Myers 1949; McDowall 1988). Diadromy has particularly strong influences on the distribution and intraspecific genetic differentiation of freshwater-breeding organisms, given the restrictions imposed by freshwater habitation on movement: diadromous species typically have much broader geographic distributions compared with purely freshwater relatives (McDowall 2001), and lower intraspecific population genetic differentiation (Allibone and Wallis 1993; Ward et al. 1994; McDowall 1999). Furthermore, population sizes in diadromous species are also expected to be larger, leading to reduced extinction rates. In a recent analysis of actinopterygian fishes (7822 species), Corush (2019) observed a positive relationship between diadromy status and diversification, using State-dependent Speciation Extinction models (Beaulieu and O'Meara 2016). In contrast, while analyzing the same taxa, Tedesco et al. (2017) obtained the opposite result, using method-of-moment estimators of diversification rate for each family. However, the vast depth and breadth of lineages surveyed in both studies increases the potential that other unconsidered evolutionary factors could confound inferences regarding diadromy (de Queiroz 2002; Donoghue and Sanderson 2015; Beaulieu and O'Meara 2016), especially considering the low frequency of this life history in the dataset (2.3%; Davis et al. 2013). Therefore, comparisons involving closely related taxa that differ primarily in diadromy status are also desirable. One such opportunity is presented by the galaxiid fishes.

Galaxiids (family Galaxiidae) represent a dominant and widespread radiation of temperate Southern Hemisphere freshwater fishes, comprising approximately 80 mostly fluvial species, of which an unusually large proportion (14) exhibit diadromy (McDowall 1987). The majority of diadromous galaxiids are amphidromous, with larvae migrating to sea and then returning to freshwater as juveniles (McDowall 2007). Across the Galaxiidae, loss of diadromy appears common, with several freshwater-limited taxa representing the product of "landlocking" from diadromous ancestors (Ovenden et al. 1993; Waters et al. 2010; Burridge et al. 2012). Diadromy is also facultative in some species, simultaneously possessing landlocked and diadromous populations (Augspurger et al. 2017) or polymorphic life histories within populations (Hicks et al. 2017). Species distributions and intraspecific genetic differentiation follow expected patterns based on life history: diadromous galaxiids are more widespread (McDowall 2001) and exhibit lower genetic differentiation among populations (Allibone and Wallis 1993; Waters et al. 2000) than nondiadromous galaxiids. The same pattern holds with respect to diadromous versus nondiadromous populations within facultatively diadromous species (Ovenden and White 1990; Waters et al. 2002; Delgado et al. 2019). Genetic differentiation among and within nondiadromous lineages is often substantially influenced by geomorphological and climatic processes (Waters et al., 2001, 2010, 2015, 2019; Burridge et al., 2006, 2007; Craw et al., 2012, 2016a,b).

In this study, we tested whether large-scale differences in mobility, represented by presence or absence of diadromy in galaxiid fishes, explain rates of diversification. Galaxiids are an ideal model to address this question as most occupy similar riverine environments as adults, with relatively few ecological differences that could confound comparisons based on dispersal ability. Furthermore, they comprise a large (approximately 80 species) group in which both migratory and nonmigratory life histories are well represented. We exploited State-dependent Speciation Extinction models to analyze a global time-calibrated phylogeny of galaxiids to elucidate the extent to which dispersal ability influences rates of diversification. Finally, we seek to increase the understanding of the conflicting migration-diversification relationships observed among diverse studies.

Materials and Methods

Phylogenetic analyses incorporated as many galaxiid species as possible. Although studies of diversification rates can accommodate missing taxa by incorporating their proportion during analysis, they assume the missing taxa are randomly distributed across the tree (Beaulieu and O'Meara 2016). However, the previously most comprehensive phylogenetic analysis of Galaxiidae (Burridge et al. 2012) lacks representation within two groups of morphologically cryptic species. Therefore, these groups were incorporated into the existing DNA dataset (599 bp *lrRNA*, 1154 bp cyt *b*, 1695 bp *RAG-1*, and 1083 bp *S7*) and reanalyzed to construct a time-calibrated phylogeny, following the same approach as previously (Burridge et al. 2012). First, the South African *Galaxias zebratus* clearly represents a group of morphologically cryptic species (Waters and Cambray 1997; Chakona et al. 2013). Cytochrome *b* data were added to represent the "nebula," "Heuningnes," "Klein," "Joubertina," "Riviersonderend," "Goukou," "Breede," "rectognathus," "slender," and "mollis" putative species (Waters and Cambray 1997; Chakona et al. 2013). Another distinct lineage previously identified from the Olifants River is also retained (Waters and Cambray 1997). Second, the Australian mountain galaxias complex (*Galaxias olidus s.l.*) is now recognized as a radiation comprising at least 15 species (Adams et al. 2014; Raadik 2014). Although not all species within the mountain galaxias complex are reciprocally monophyletic for DNA variation (Raadik 2014), we chose cyt *b* sequences that represent a lineage distinct for each species. Although the molecular phylogeny reconstructed for these recently diverged mountain galaxias may not represent those of the actual species, all species exhibit the same life history and the chronology of their molecular divergence is very shallow. Following Adams et al. (2014) and Raadik (2014), the previously analyzed data ascribed to *G. olidus* now corresponds to *Galaxias oliros*. In addition to data from these two recently recognized complexes, cyt *b*, *lrRNA*, and *S7* data from a new species *Galaxiella toourtkoourt* (Coleman et al. 2015) were also included. It has become apparent that previously analyzed sequence data considered to represent *Aplochiton taeniatus* actually represent *Aplochiton zebra*, and therefore new cyt *b* data have been substituted. Homologous data from the recently resurrected diadromous *Aplochiton marinus* were not available at the time of analysis, representing the only excluded known galaxiid species. All new data analyzed relative to Burridge et al. (2012) are summarized in Table S1. The proportion of missing species was set at 0.07 for diadromous species (1/14) and 0.03 (2/68) for nondiadromous species (estimating that two nondiadromous species may still not be recognized within *G. zebratus s.l.*, *G. olidus s.l.*, or elsewhere).

The molecular phylogeny of the Galaxiidae, with branches scaled proportionally to evolutionary time, was obtained following the approach of Burridge et al. (2012). The analysis used a relaxed molecular clock and included nine biogeographic age calibration points within the family, together with nine fossil calibration points from outgroup lineages. BEAST 2.6.2 (Bouckaert et al. 2019) postburnin run exceeded 6.9×10^8 steps, with stationarity and convergence assessed using Tracer (Rambaut et al. 2018). A total of 35,586 trees were retained for the construction of a maximum clade credibility tree using TreeAnnotator. Outgroups and lineages that represent intraspecific galaxiid divergences were pruned such that each galaxiid species was represented by a single lineage, with two exceptions. First, the diadromous *Galaxias brevipinnis* is resolved as polyphyletic, with New Zealand and Australian clades exhibiting molecular divergence consistent with interspecific differentiation elsewhere in the family (Waters et al. 2010; Burridge et al. 2012). Second, *Galaxias maculatus* exhibits substantial molecular divergence among Australian, New Zealand, and South American populations and arguably represents three species (Waters and Burridge 1999). *Galaxias niger* was excluded as this appears to be synonymous with Australian *G. brevipinnis* based on the depth of molecular divergence and morphological plasticity of the latter.

Taxa were coded as diadromous or nondiadromous based on knowledge derived from published accounts of species biology (largely summarized by Augspurger et al. 2017). Observations of migratory status of *Aplochiton* spp. indicate that populations of *A. taeniatus* might variously be freshwater resident or diadromous (catadromous or amphidromous), and some landlocked populations are known also for *A. zebra* (Alò et al. 2019). The widespread distribution of the latter (including the Falkland Islands/Islas Malvinas), however, implies a recent diadromous history, and therefore this species was considered to possess both life histories. Semi-anadromy is observed in *Lovettia* (Schmidt et al. 2014), the sister-lineage of *Aplochiton*. The majority

of diadromous species also exhibit landlocked populations (*Aplochiton* spp., *G. brevipinnis*—Australia and New Zealand, *Galaxias maculatus*, *Galaxias truttaceus*, *Galaxias argenteus*, and *Galaxias fasciatus*) and were treated in two ways. First, they were coded as diadromous, because their landlocked populations are likely recent derivatives of diadromous populations, which better reflects their state through evolutionary time (Ovenden et al. 1993; Waters et al. 2010). Second, we also tested models under which species could occupy multiple states simultaneously (using GeoHiSSE, below).

Analyses were conducted under the Hidden-state Speciation Extinction (HiSSE) framework (Beaulieu and O'Meara 2016) to test the null hypothesis of no relationship between diversification rate and migratory status. This approach avoids potential errors associated with the use of "equal rate" null hypotheses (Rabosky and Goldberg 2015), and also accommodates shifts in diversification that could also be related to states that are unobserved or "hidden", to reduce inferential errors with respect to migratory status (Beaulieu and O'Meara 2016). All analyses were performed in R using the *hisse* (Beaulieu and O'Meara 2016) and *diversitree* (FitzJohn 2012) packages. The parameters of the HiSSE model are rates of transition among character states (diadromy, hidden state: presence/absence), and species turnover (speciation + extinction) and extinction fraction (extinction/speciation). State transition rates were either equal or unequal. Simultaneous changes in migration status and the hidden state were excluded. The null model assumed that species turnover and extinction fraction did not vary with respect to migratory status, but did with respect to hidden state ("A" or "B"). The null was then compared to a model lacking hidden states, where species turnover and extinction fraction were allowed to vary with migratory status. Models were compared based on Akaike information criterion (AIC). Additional models were constructed to test whether the presence of hidden states within either diadromous or nondiadromous lineages could explain an apparent difference in diversification with respect to migratory status (Beaulieu and O'Meara 2016). A contribution of a nested hidden state to diversification might reveal that additional characteristics are also important in explaining diversification rates (Beaulieu and O'Meara 2016).

Models were also conducted under the GeoHiSSE framework (Caetano et al. 2018), originally developed for biogeographic analysis where species can occupy multiple areas, but here applied to accommodate those species exhibiting both diadromous and nondiadromous populations (see list above). Analyses analogous to HiSSE were performed using the *hisse* package. These analyses allowed transitions directly between diadromy and nondiadromy (no requirement of a polymorphic intermediate state; "include.jumps = TRUE") and also from a polymorphic state to either of the individual states ("separate.extirpation = TRUE").

Results

The time-calibrated molecular phylogeny for galaxiids is presented in Figure 1. This tree confirms the combined monophyly of recently recognized species-rich complexes (*G. zebratus* and *G. olidus* groups) that have not previously been included in comprehensive phylogenies of the family. Extant lineages of the former complex (South Africa) are suggested to have diversified over the last ~20 million years, whereas the latter (Australia) evolved more recently (~10 million years ago). Our tree was broadly well-resolved, with topological uncertainty generally restricted to shallow nodes. Furthermore, only 14% of nodes with low support values (<0.95) could potentially influence placement of diadromous species, and again, mostly at shallow levels.

Some models allowing different rates of diversification between diadromous and nondiadromous lineages received substantially more support than the null model, which allowed diversification to vary only with respect to a hidden state (Table 1). When allowed to differ with migratory status, rates of speciation and extinction were higher in the nondiadromous lineages compared with diadromous, as were rates of diversification, with the exception of extinction in one model (Fig. 2). The optimal model accommodated a hidden state within nondiadromous lineages and equal-state transition rates, although a model with higher state transition rates into diadromy had only slightly lower support (Table 1). The GeoHiSSE analysis, enabling recognition of both states within a lineage ("polymorphic": diadromous and non-diadromous simultaneously), produced similar results to the HiSSE analysis. The best GeoHiSSE model accommodated a hidden state within nondiadromous lineages, and equal state transition rates, although here the related model with unique transition rates was substantially inferior (Table S2); all other models received much lower support. When allowed to vary with migratory status, GeoHiSSE rates of diversification were higher in the nondiadromous relative to diadromous lineages, and similar trends were found for speciation and extinction rates (Fig. S1).

Under the best HiSSE model, which employed a hidden state within nondiadromous lineages, faster diversification rates at tree tips are associated more highly with hidden state "B", which appears linked with recency of divergence from diadromous lineages across the tree, with the exception of the *G. olidus* complex (Fig. 3). Specifically, those nondiadromous tree tips that are closely related to *G. maculatus*, *G. brevipinnis* (Australian and New Zealand), *G. truttaceus*, or *Galaxias postvectis* exhibit higher rates of diversification than those not closely related to diadromous lineages, excluding the *G. olidus* complex (Fig. 3). Although *tip* diversification rates were not substantially different with respect to the hidden state within nondiadromous lineages (~0.090–0.145), *across the tree* these states differed almost twofold, with greater variation in their underlying speciation and extinction rates, with faster rates in hidden state "B" (Fig. 2).

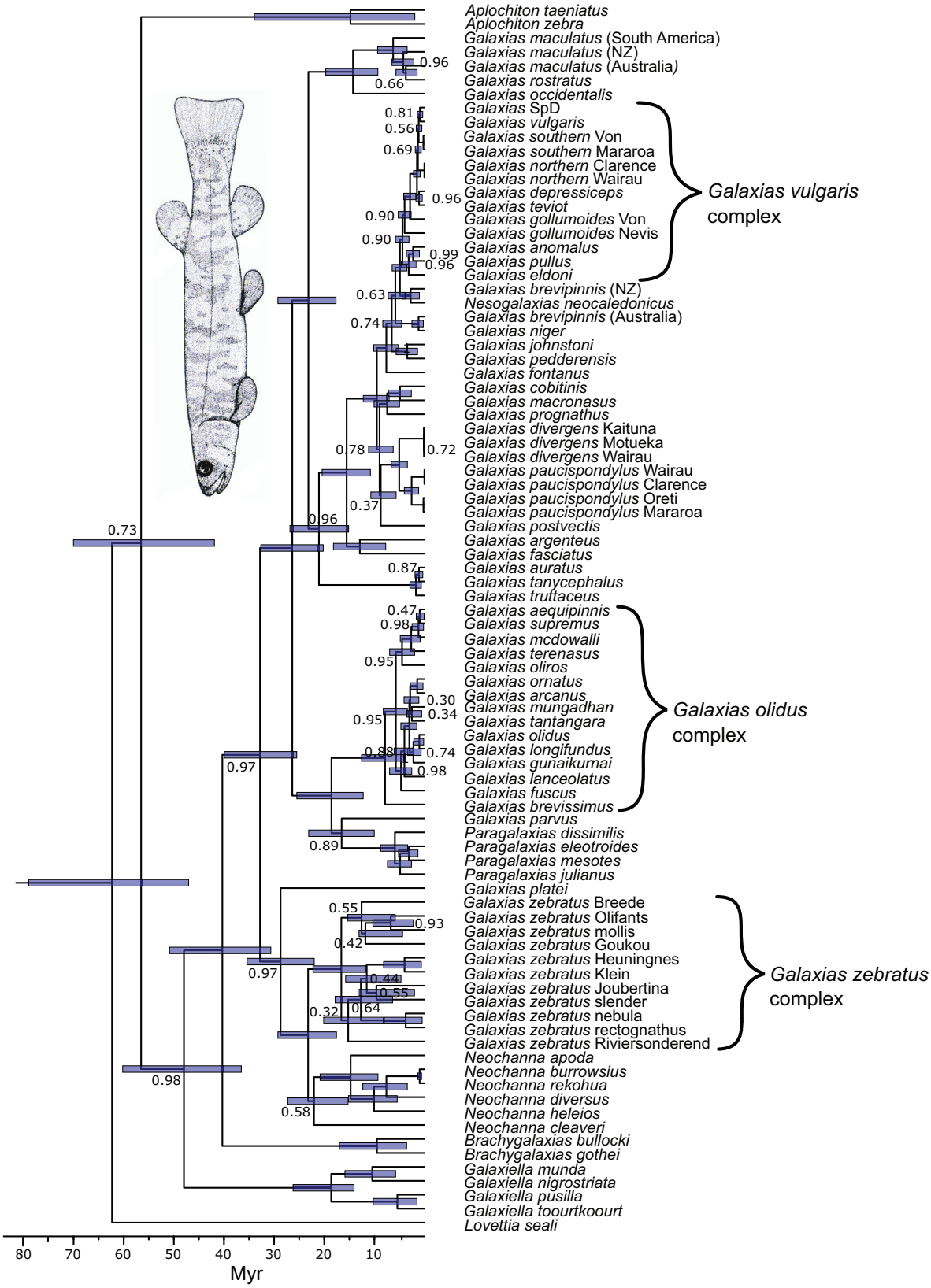

Figure 1. Bayesian estimate of phylogeny for the Galaxiidae. Branches are scaled according to posterior estimates of divergence times. Horizontal bars at nodes indicate the 95% posterior probability of the age of that node. Numbers represent posterior probabilities for nodes when less than 1.00. Outgroups are not illustrated. Some galaxiid tips were pruned prior to State-dependent Speciation Extinction (SSE) analysis, as described in the main text. Illustration depicts *Galaxias vulgaris*.

Table 1. HiSSE models of diversification and state transition rates with respect to diadromy status (diadromous and nondiadromous) in galaxiid fishes. *k*, total number of parameters in the model.

Model	*k*	logL	AIC	ΔAIC	*ω*	q01	q10
1. Null model; hidden state; equal state transitions	5	−290.00	590.00	5.45	0.03	1.54	1.54
2. Unique rates; hidden state; unequal state transitions	6	−290.35	592.70	8.16	0.01	1.52	1.54
3. Unique rates; hidden state; equal state transitions	5	−290.36	590.72	6.18	0.02	1.55	1.55
4. Unique rates; hidden state within non-diadromous; unequal state transitions	10	−282.46	584.92	0.38	0.42	3.93	0.00
5. Unique rates; hidden state within non-diadromous; equal state transitions	7	−285.27	584.54	0.00	0.51	3.17	3.17
6. Unique rates; hidden state within diadromous; unequal state transitions	10	−288.63	597.26	12.71	0.00	5.52	44.14
7. Unique rates; hidden state within diadromous; equal state transitions	7	−289.91	593.82	9.28	0.00	1.94	1.94

ω = AIC weights. Rates of transition between diadromous and nondiadromous states were either equal or unequal, with q01 representing transitions from nondiadromous to diadromous, and q10 the opposite. "Unique rates" allowed turnover and extinction fraction to vary with respect to states. Use of a "hidden state" accommodates variation in turnover and extinction fraction with respect to an anonymous trait that may be distributed across all taxa, or within a particular life history category.

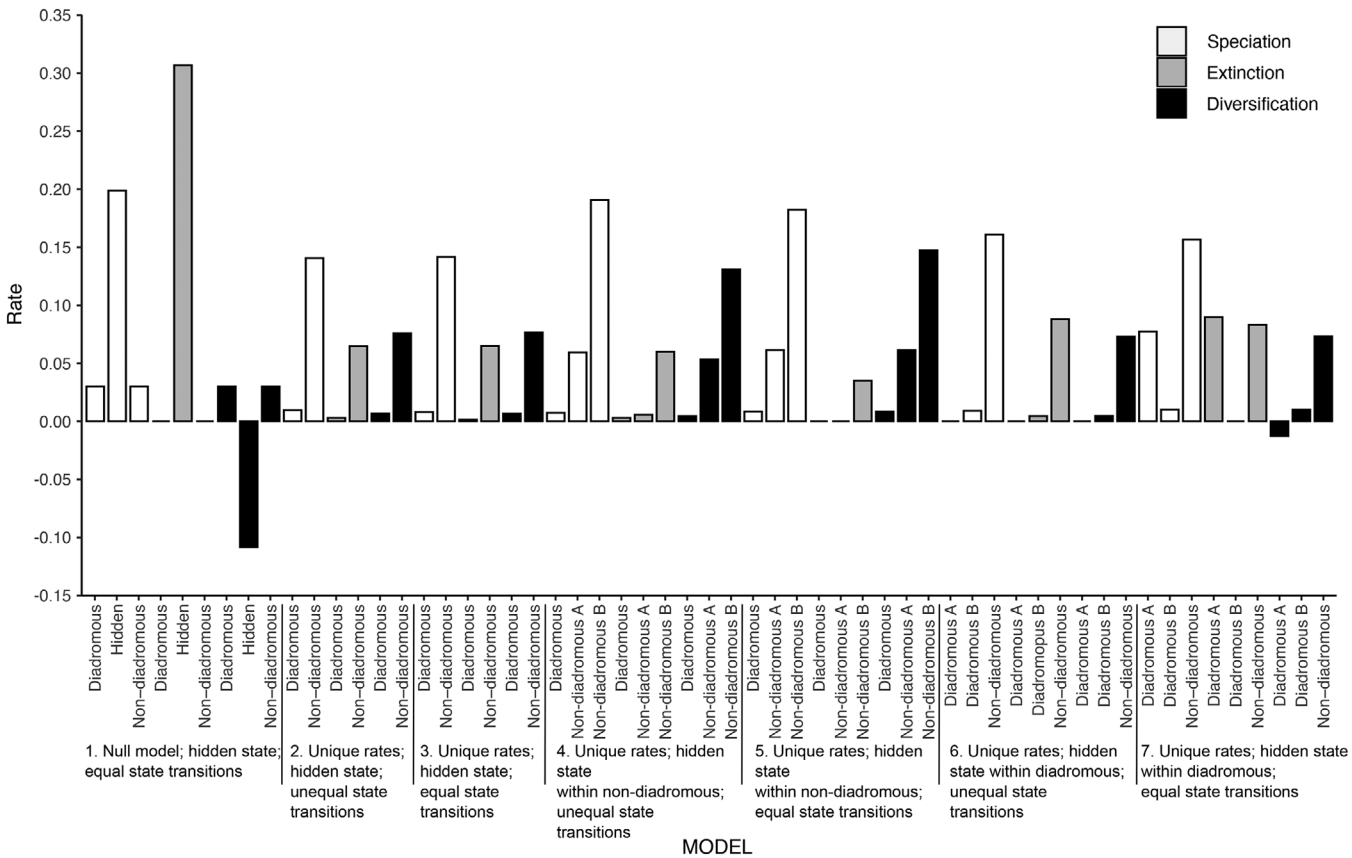

Figure 2. Rates of speciation, extinction, and diversification among diadromous and nondiadromous lineages when analyzed under HiSSE models employing different heterogeneity in their rates, and either equal or unequal transition rates between states. Model numbers correspond to Table 1.

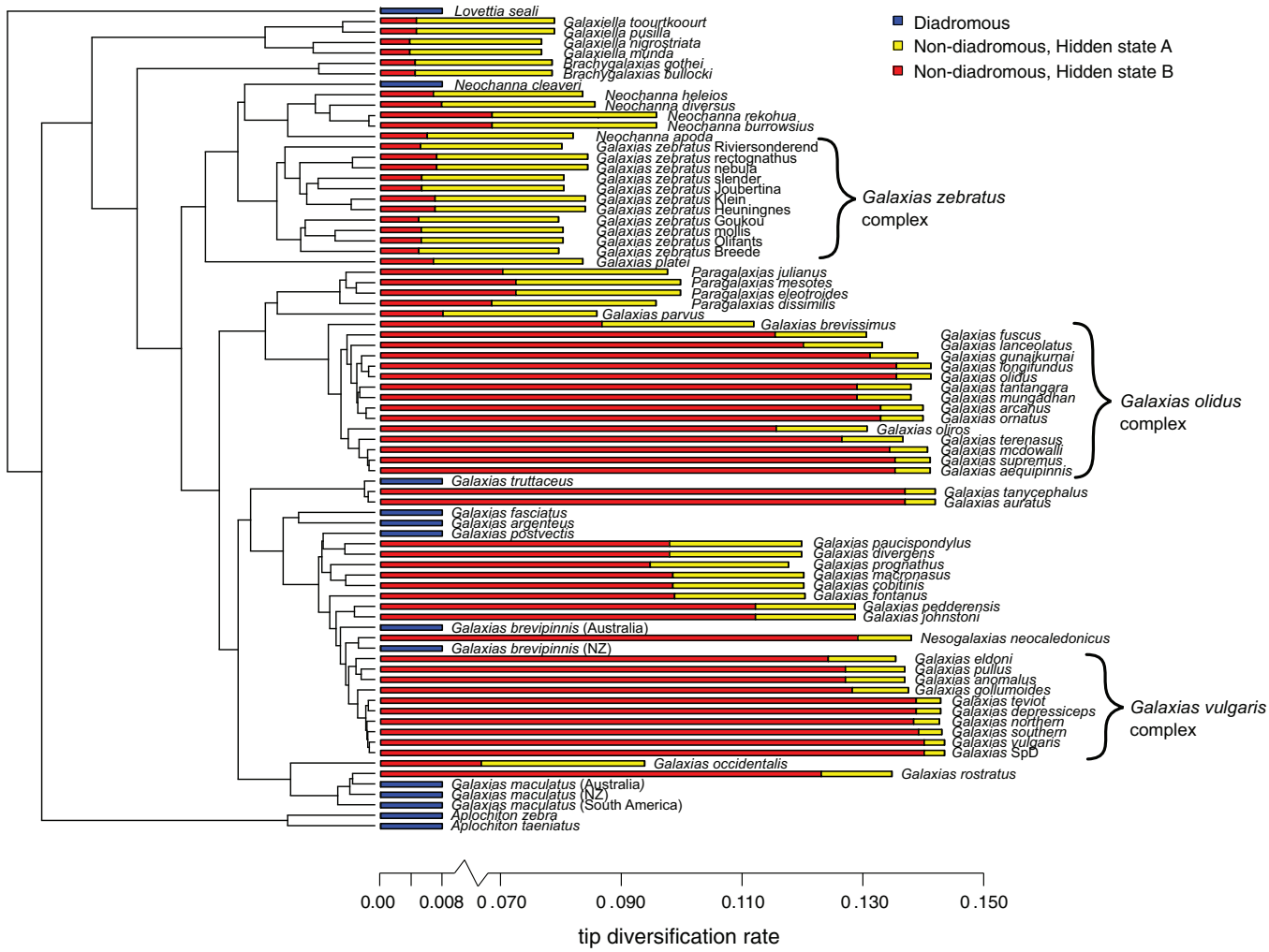

Figure 3. Rates of diversification at the tips of the galaxiid phylogenetic tree under a HiSSE model employing a hidden state within nondiadromous lineages and equal transition rates among states (diadromous and nondiadromous). The assignment of tips to states is indicated by the ratio of colors within bars, regardless of bar length, which represents only diversification rate (note scale break).

Discussion

The major finding of this study was the presence of higher diversification rates in nonmigratory than migratory galaxiids. Although estimated extinction rates were higher in nonmigratory galaxiids, consistent with expectations from theory and some empirical tests (Rolland et al. 2014; Corush 2019), they were not sufficiently elevated to suppress overall diversification rates. Although our findings are broadly consistent with a wide variety of analyses linking high diversification rates with "dispersal reduction" (Waters et al. 2020), including one study of migratory behavior in fishes (Tedesco et al. 2017), they directly contrast with the conclusions of some recent empirical studies that suggested migratory behavior leads to higher diversification (Rolland et al. 2014; Fuchs et al. 2015; Corush 2019; Gómez-Bahamón et al. 2020) or has no effect (Kennedy et al. 2016). The lower diversification rates for migratory galaxiids could be explained by lower natal philopatry relative to other migratory taxa, perhaps reflecting either unpredictability of freshwater habitat quality—and hence greater benefit of straying—or lower locomotory abilities of the migratory stage in galaxiids: larvae. Alternatively, discrepancies in migration-diversification relationships observed among studies more generally might be explained by the spatial context of study systems relative to species dispersal abilities.

Previous research has suggested higher species richness and diversification manifests at intermediate dispersal abilities (the "intermediate dispersal" model), enabling colonization of new areas, but then subsequent divergence under limited gene flow (Claramunt et al. 2012; Ashby et al. 2020). However, the categorization of "intermediate dispersal" is potentially influenced by the spatial context of a study system relative to dispersal capabilities (Kisel and Barraclough 2010; Claramunt et al. 2012). For example, diadromous galaxiids appear to possess

strong dispersal capabilities; their "whitebait" larval duration extends up to 23 weeks (McDowall 1987), and they have been observed 700 km offshore (McDowall et al. 1975). Overall, dispersal has facilitated widespread temperate distributions by diadromous galaxiids within landmasses, but also appears to have precluded population genetic divergence (Ovenden and White 1990; Waters et al., 2000, 2002). At the same time, the dispersal kernel of diadromous galaxiids may provide too few propagules for the frequent colonization of distant landmasses (i.e., other continents) in the face of any pre-existing competitors (Waters et al. 2013). Thus, the disjunct spatial distribution of temperate freshwater habitats in the Southern Hemisphere (spatial context), combined with dispersal abilities of diadromous galaxiids, may explain their comparatively low diversification rate. Under this framework, the contrasting migration-diversification relationship observed by Corush (2019) could be explained by the predominantly tropical and temperate Northern Hemisphere distributions of most diadromous taxa, where marine distances among freshwater environments are more continuously distributed, enabling "intermediate dispersal" scenarios to be more readily realized by migratory than nonmigratory lineages. Therefore, discrepancies in dispersal-diversification relationships among studies might be explained by differences in their spatial context relative to species dispersal capabilities, under the perspective of the "intermediate dispersal" model.

Although our analyses indicated overall greater diversification in nondiadromous than diadromous lineages, substantially more support was obtained when including the presence of hidden states that might explain variation in diversification among nondiadromous lineages. These hidden states differed in diversification rates more subtly from one another than relative to diadromous lineages. Although carrying less biological significance than the diadromous/nondiadromous dichotomy, these hidden states are still worthy of consideration to understand why overall rates differ with migratory status. Elevated diversification rates are evident in nondiadromous lineages that have only recently descended from ancestors with diadromous species. This includes the *Galaxias vulgaris* complex (∼10 species) in New Zealand, *Galaxias auratus* and *Galaxias tanycephalus* in Tasmania, and *Galaxias rostratus* in mainland Australia. Additionally, the New Zealand "pencil" galaxiids (*Galaxias paucispondylus*, *Galaxias divergens*, *Galaxias cobitinis*, *Galaxias prognathous*, and *Galaxias macronasus*) and Tasmanian *Galaxias fontanus*, *Galaxias pedderensis*, and *Galaxias johnstoni* are somewhat closely related to *G. postvectis*, and their mid-Miocene divergence across the Tasman Sea also implies a diadromous history. These elevated rates may thus be a consequence of "landlocking," the adaptation of diadromous taxa to a purely freshwater lifecycle (McDowall 2001).

Landlocking has been proposed as a major source of diversity in freshwater fishes, based on observations within closely related taxa (McDowall, 1988, 2001; Palkovacs et al. 2008; Bracken et al. 2015; Augspurger et al. 2017), including galaxiids (Ovenden et al. 1993; Waters and Wallis 2001), but also based on an analysis across ray-finned fishes (Corush 2019). In the case of galaxiid fishes, however, it appears that diversification subsequent to landlocking may be particularly important, perhaps more so than the number of landlocking events per se. For instance, a detailed analysis of the *G. vulgaris* complex in New Zealand highlights diversification subsequent to landlocking of a *G. brevipinnis*-like ancestor (Waters et al. 2010). Similarly, high diversification rates recovered in clades that have no recent history of diadromy suggest that loss of migration is a precursor rather than driver of diversification (e.g., the *G. olidus* complex). More broadly, across our entire phylogeny symmetrical rates of state transition were either clearly favored over asymmetrical rates (GeoHiSSE), or when equivocal (HiSSE), asymmetry reflected transitions to, rather than from, diadromy.

Elevated diversification rates among some nondiadromous lineages may be best explained by geomorphological changes to dendritic networks (e.g., river capture and river piracy), which appears an important source of cladogenesis in New Zealand galaxiids at levels of both species (Craw et al., 2012, 2016a,b; Waters et al. 2015) and populations (Waters et al. 2001; Burridge et al., 2006, 2007). Similarly, Tasmania is the only part of Australia to have experienced extensive glaciation during the Pleistocene, and Tasmanian nondiadromous species with elevated diversification rates such as *G. pedderensis* and *G. johnstoni* occupy areas proximate to past glacial influence (Colhoun et al. 1996). Therefore, exposure to geomorphological processes facilitating the allopatric isolation of populations may promote speciation. Furthermore, although extinction rates should be generally higher in nondiadromous than diadromous lineages, geological events such as river capture can result in range expansions of newly isolated lineages (Burridge et al. 2006), such that their extinction rates may actually be reduced relative to background freshwater rates, and additionally by the low diversity of other species and reduced competition in these areas (Betancur-R. et al. 2012). Therefore, drainage evolution and other abiotic processes may explain the elevated rates observed for numerous nonmigratory galaxiid lineages.

The other main group with strong association to the hidden state with higher rates of speciation, extinction, and diversification is the *G. olidus* complex, comprising at least 15 species. However, this group has neither recent ancestry with an extant diadromous lineage nor occupies regions of particularly dynamic geomorphological history (but see Holdgate et al. 2008; Waters et al. 2019). The majority of the representatives of the

G. olidus complex are narrowly distributed, with only four more-widespread species showing genetic evidence for recent intra- or intercatchment dispersal (Adams et al. 2014). The high rates of diversification in this group may stem from their apparent ability to disperse upstream and across barriers. Climbing ability of some galaxiids is facilitated by raised fin ray lamellae on paired fins (pectoral and pelvic), which notably enable *G. brevipinnis* to traverse waterfalls of 60 m—by providing a larger ventral surface area for adhesion to substrates (McDowall 2003). Similar morphological adaptations for occasional dispersal across significant intra- or even intercatchment barriers may have promoted diversification in the *G. olidus* complex by the establishment of allopatric distributions, despite the comparative geomorphological stability of its landscapes (Raadik 2014). The *G. olidus* complex contrasts particularly with the South African *G. zebratus* complex, which also exhibits morphologically cryptic diversity, but lacks raised fin lamellae, and is not found above major instream barriers, and hence diversification rates are apparently slower.

Conclusions

Overall, we infer that opportunities afforded by geological history or morphological adaptations have enabled nonmigratory galaxiid lineages to disperse between otherwise physically isolated habitats, and to experience greater rates of diversification than migratory lineages. Divergence of migratory lineages has apparently been inhibited by dispersal capabilities that genetically homogenize populations within landmasses, yet have been insufficient for the frequent colonization of distant landmasses. The main potential caveat to our study is that which usually applies to studies employing model selection approaches: although we find support for migratory status influencing diversification rate, if diversification is more strongly influenced by some other trait that is not in our candidate set of models, our inferences may be misleading (Beaulieu and O'Meara 2016). Additionally, the small number of tree tips and bias in character state ratio in our study system also presents a need for caution (Davis et al. 2013), but here at least we minimized this error by sampling in excess of 95% of the Galaxiidae, a group of approximately 80 species in which both life histories are well represented (McDowall 1987). Investigations across greater taxon breadth would increase bias in character-state ratio and also entail greater phylogenetic depth, which increases the potential that other unconsidered traits (e.g., distribution) could confound inferences involving diadromy. We encourage future investigations of the potential interaction of dispersal ability (accurately measured) and the proximity of suitable habitats on diversification rates, in the context of the "intermediate dispersal" model. More generally, galaxiid fishes also provide a key opportunity to investigate the molecular mechanisms underlining transitions between diadromous and nondiadromous life histories, and whether the multiple transitions observed among galaxiids reflect identical mechanisms, both relative to one another, and to other fishes that have already been investigated in detail (Jones et al. 2012; Michalak et al. 2014; Terekhanova et al. 2014; Lemopoulos et al. 2018; Delgado et al. 2019; Waters et al. 2020).

AUTHOR CONTRIBUTIONS

CPB and JMW conceptualized the study. CPB analyzed the data and presented the results. CPB wrote the original draft. JMW reviewed and edited the manuscript. JMW acquired funding for this study.

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DATA ARCHIVING

Data and code used in this study are available on Dryad (https://doi.org/10.5061/dryad.zkh18937n).

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1. New data analyzed relative to Burridge et al. (2012).

Table S2. GeoHiSSE models of diversification with respect to diadromy status (diadromous, nondiadromous, polymorphic) in galaxiid fishes. *k*, total number of parameters in the model.

Figure S1. Rates of speciation, extinction, and diversification among diadromous, non-diadromous, and polymorphic (diadromous and nondiadromous) lineages when analyzed under GeoHiSSE models employing different heterogeneity in their rates, and either equal or unequal transition rates between states.